25900462 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCIAM200 - EMBEDDED SYSTEM PROGRAMMING TOOLS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. What is the result of the expression 2 ** 3 in Python? [K] / [CO1]

   a). 6 b). 8

   c). 9 d). Error

2. Which Python library is commonly used to handle images (pictures)? [K] / [CO2]

   a). numpy b). PIL / Pillow

   c). matplotlib d). pygame.mixer

3. Which geometry manager is best for tables? [U] / [CO3]

   a). pack b). grid

   c). place d). align

4. If x = 5 and y = 5.0, what is the difference between them in Python? [U] / [CO1]

a). No difference, both are integers
b). x is an integer and y is a floating-point number

c). Both are floating-point numbers
d). x is a string and y is an integer

5. Why do we often convert strings into numbers in Python applications? [U] / [CO2]

a). To perform arithmetic operations on input data.
b). To store values in memory more efficiently.

c). To increase program speed automatically.
d). To avoid string concatenation errors.

6. Tkinter is written in? [K] / [CO3]

a). C
b). C++

c). Python
d). Java

7. Which of the following statements explains why we use string formatting in number system applications? [U] / [CO2]

a). To change numbers into strings only.
b). To easily represent numbers in different bases like binary, octal, or hex.

c). To increase execution speed.
d). To remove errors in numeric operations.

8. How many geometry managers does Tkinter have? [K] / [CO3]

a). 2
b). 3

c). 4
d). 5

9. Why is it necessary to configure Wi-Fi settings during the first boot of Raspberry Pi 4? [U] / [CO1]

a). To activate HDMI ports
b). To enable remote access and internet connectivity

c). To increase processing speed
d). To update the GPIO pins

10. Which geometry manager provides most control? [U] / [CO3]

a). pack
b). grid

c). place
d). align

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. List the main numerical data types and sequence types in Python (int, float, complex, string, list, tuple, range) and give one suitable example for each. [K] / [CO1]

12. Write a Python program to perform a digital read from a GPIO pin connected to a push button. Explain how the input value is returned. [U] / [CO2]

13. Which method is used to start the Tkinter event loop? [K] / [CO3]

14. Describe the purpose of the functions GPIO.setup() and GPIO.cleanup(). Give a suitable code snippet to illustrate their use. [U] / [CO2]

15. What does GUI stand for? [K] / [CO3]

16. List the two modes of GPIO operation (Input and Output) and explain the difference between them. [K] / [CO2]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Explain the port layout of Raspberry Pi 4 with the help of a neat diagram. Discuss the role of USB ports, HDMI ports, Ethernet, and GPIO pins. [U] / [CO1]

18. Explain the working of Label, Button, Entry, and Checkbutton widgets with examples. [K] / [CO3]

19. With the help of a suitable example, demonstrate the basic flow of a Python program including input, processing, and output statements. [U] / [CO1]

20. Discuss how control statements help in decision-making and iteration in Python programs. Write a program that uses at least one conditional statement and one loop, and explain its output. [U] / [CO2]

[10x2 = 20]